

Electronic Signature Information

Name	DOC0757335
Revision	6
Type	Controlled Document
Title	Direction 5371824_ 6Ch Flex Coil Service Manual
Reason For Change	Updated the service manual to remove Configuration CD and Service Pack DVD.
Originator	212027361_dale_b_seaman
Release Date	02/13/2024 02:15:32 PM
Obsolete Date	

File Name	File Description	File Size (Bytes)
5371824_Rev4_6ChannelFlexCoilServiceManual_DOC0757335.pdf	6Ch Flex Coil Service Manual	1545001

Route	Signer	Function	Status	Comments	Completion Date
R-10972553	212479745_mervat__mustafa	QA	Approve	Approved as a Task Assignee	10 Feb 2024 18:00:03 GMT
R-10972553	212027704_aleksey__zemskov	LSD	Approve	Approved as a Task Assignee	30 Jan 2024 16:44:55 GMT
R-10972553	212027673_keith__richmond	LSI - Program Manager	Approve	Approved as a Task Assignee	13 Feb 2024 14:07:42 GMT
R-10972553	212036283_rebecca_t_david	Sr. Tech Project Manager	Approve	Approved as a Task Assignee	30 Jan 2024 17:38:16 GMT
R-10972553	212019387_emad__abu tabanjeh	Coil Integration Architect	Approve	Approved as a Task Assignee	30 Jan 2024 14:51:06 GMT
R-10972553	223043750_canyon__aspy	Engineer	Approve	Approved as a Task Assignee	5 Feb 2024 22:24:31 GMT

Periodic Review

There are no signatures or routes related to this business object.

Obsolescence Approval

There are no signatures or routes related to this business object.

* Printed versions are For Reference Only *

+ Indicates a task was reassigned from an original assignee



GE HealthCare

SIGNA Artist, MR450, MR450w, MR450w GEM, SIGNA Architect, MR750, MR750w GEM

1.5T and 3.0T 6-Channel Flex Coil Service Manual

5371824
Revision 4

General service documentation.
© 2010-2023 GE HealthCare. GE is a trademark of General Electric Company used under trademark license.
Reproduction and/or distribution is prohibited.

Released

Language Policy

DOC0371395 - Global Language Procedure

ПРЕДУПРЕЖ ДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. - Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. - Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ÆÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploataavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploataavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutra idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCIÓN (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyonu bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

ЗАСТЕРЕЖЕННЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. • Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. • Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. • Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.
----------------------	---

Contents

Chapter 1 Getting started	11
1.1 Introduction.....	11
1.1.1 Shipping components.....	11
1.1.2 Damage in transportation.....	11
1.1.3 Product locator	12
1.1.4 Requirements and specifications.....	13
1.1.5 Coil description	13
1.1.6 Modes of operation	15
Chapter 2 Installation	17
2.1 Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance (MCQA) test	17
Chapter 3 Replacements	21
3.1 Field Replaceable Units (FRUs).....	21
3.2 Replacing the 6-channel flexible phased array coil assembly	22
3.2.1 Replacing the 6-channel flex phased array coil half.....	22
3.2.2 Replacing the Quick Disconnect (QD) assembly.....	23
3.2.3 Replacing the Quick Disconnect (QD) assembly pre-amplifiers	24
Chapter 4 Troubleshooting	29
4.1 Troubleshooting the 6-channel flex coil.....	29
4.1.1 Troubleshooting the 6-channel flex coil using the multi-coil bias diagnostic	31
Revision history	35

Page intentionally left blank

Chapter 1 Getting started

1.1 Introduction

The 1.5T and 3.0T 6-channel flexible phased array coil (6-channel flex coil) is compatible with the GE HealthCare 1.5T MR450, MR450w Base, MR450w GEM, and the GE HealthCare 3.0T MR750, MR750w, and MR750w GEM whole body MRI systems. A P to A adapter enables use of the 6-channel flex coil on the SIGNA Artist and SIGNA Architect systems, which do not have an A-port.

The 6-channel flex coil is a receive only, multi-coil array designed to give the optimal Signal-to-Noise Ratio (SNR) and uniformity for high resolution examinations. The coil enables the imaging of multi-oblique slices of the head and other areas of the body. The multi-element design improves the coverage and resolution.

1.1.1 Shipping components

The table below includes the coil contents shipped for the specified catalog number.

Table 1-1 Shipping contents by catalog number

Description for M0049SS	Part number	Quantity
1.5T 6-Channel Flex Phased Array Coil with a Quick Disconnect Assembly	5363383	1
MR 6-Channel Phased Array Flex Coil Operator Manual	5371823	1
1.5T and 3.0T 6-Channel Flex Coil Service Manual for MR450, MR750, MR450w, GEM, Artist, Architect	5371824	1
Description for M0050SS		
3.0T 6-Channel Flex Phased Array Coil with a Quick Disconnect Assembly	5363393	1
MR 6-Channel Phased Array Flex Coil Operator Manual	5371823	1
1.5T and 3.0T 6-Channel Flex Coil Service Manual for MR450, MR750, MR450w, GEM, Artist, Architect	5371824	1
Description for M7006CW		
P to A 6-Channel Flex Coil Adapter	5771497	1

1.1.2 Damage in transportation

Examine all packages closely at delivery. If damage is apparent, do the following:

1. Make sure the notation *damage in shipment* is written on all copies of the freight or express bill before delivery is accepted or signed for by a GE HealthCare representative or a hospital receiving agent.
2. Whether noted or concealed, you must report damage to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and hold the contents and containers for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

3. Call Global Parts at 1-800-548-3366 (option 6) to file a report of the damage.
4. Contact your local service coordinator for more information on this process.

1.1.3 Product locator

NOTICE



The 6-channel flex coil catalog contains three different, serialized components. The adapter (required only for SIGNA Artist and SIGNA Architect) also includes a Product Locator Card (PLC). Make sure all information is accurate for each catalog in Siebel/CRM for proper warranty coverage and that site history is maintained.

The Global Install Base Database (also known as the Product Locator System) tracks the shipment and trans-shipment of the serialized models. When parts are shipped on an order, all serial numbers are migrated to Siebel/CRM as part of the asset creation.

During installation, the install FE will make sure that the model and serial numbers are part of the asset. Use either the gib web interface (http://gib.gehealthcare.com/gib/gib_entry.jsp) or the MyPLCReader app. This information will then be loaded directly to Siebel by the IB verification team. If components are missing, submit a Global Installation Quality (GIQ) request.

To install the MyPLCReader app, use the following procedure.

1. From the GE app icon, search for and Install the MyPLCReader app on your iPhone or iPad.

Figure 1-1 MyPLCReader app



2. Update the Configuration page.
 - 2.1. The first time the app opens, the Configuration page will open.
 - 2.2. Enter your SSO, Country and **if in US select the edit button to change the default email address** (by default, this app will send the email to the Europe IB

Team Resource.EuropeanCentralAdminGIB@ge.com). The email to be used in US is usa.ib.verification@ge.com.

3. Start your first scan.
 - 3.1. Go to the Menu, make the appropriate selection for the job you are doing: **New Install**, **Upgrade System**, or **Update System**.
 - 3.2. Enter the applicable data, install end date, system ID, and RFS number.
 - 3.3. Click **Scan Bar Codes** and start scanning each bar code on the PL Cards.
 - 3.4. Click the **Email** button.
 - 3.5. The email is sent to the default email address, which was set up earlier.

1.1.4 Requirements and specifications

Coil installation and configuration requires the services of a GE HealthCare Service Representative. Store the coil in an air-conditioned scan room or equipment room.

Table 1-2 Environmental requirements

Measurement type	Transport and storage	Operating conditions
Barometric pressure	500 to 1060 hPa	700 to 1060 hPa
Humidity	10 to 95% non-condensing	10 to 90% non-condensing
Temperature	-40 to 65°C (-40 to 149°F)	15 to 21°C (59 to 69.8°F)

Table 1-3 Coil frequency specifications

Field strength	Coil model	Frequency
1.5T	6-Channel Flex Phased Array Coil	63.86 MHz $\pm$ 50 kHz
3.0T	6-Channel Flex Phased Array Coil	127.73 MHz $\pm$ 50 kHz

1.1.5 Coil description

The 1.5T and 3.0T 6-channel flexible phased array coil (shown below) is a receive only, multi-coil array optimized for high resolution exams. The coil enables the user to image multi-oblique slices where the intended areas of interest include, but are not limited to: head, knee, shoulder, foot, wrist, and other extremities.

Figure 1-2 6-channel flexible phased array coil - identical halves



The 6-channel flexible phased array coil consists of two flexible halves that enable large openings and flexibility. The flexible halves conform to the contours of the patient and accommodate the patient's anatomy. This maximizes signal integrity and clinical utility. Both coil halves are connected to the system through an intermediate adapter connector, called a Quick Disconnect (QD) assembly shown below.

NOTE

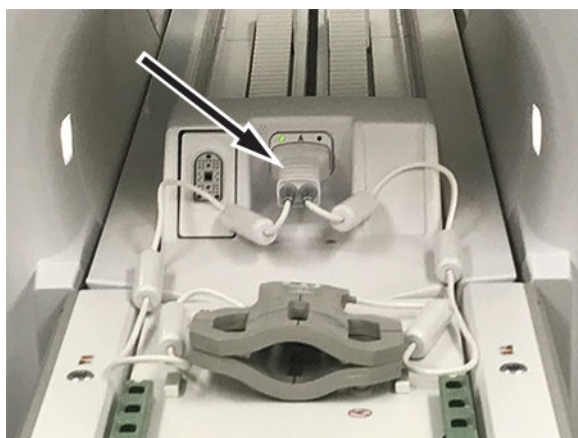
Coil setup flexibility allows the use of only one coil half, if desired.

Figure 1-3 QD assembly

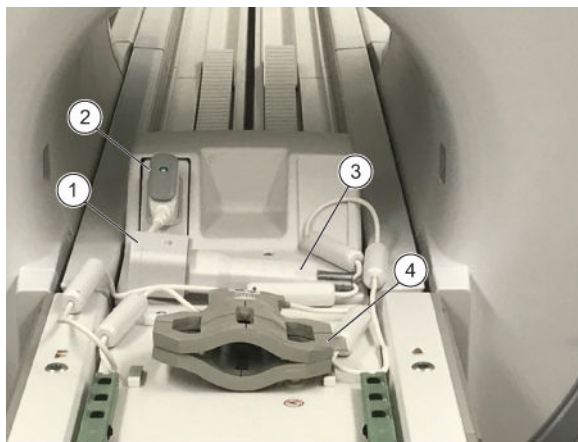
**NOTE**

The QD assembly connects directly to the LPCA of the MRI system shown below.

Figure 1-4 6-channel phased array flex coil connector



For systems without an A-port, the P to A adapter is required.

Figure 1-5 6-channel phased array flex coil connector with adapter plugged into a P-port

Item	Description	Item	Description
1	Adapter box	3	QD box
2	P-connector	4	6-channel flex coil

1.1.6 Modes of operation

The 6-channel flex coil can be operated in multiple modes. Coil modes for neuro applications:

- 6ch Neuro Flex: Both coil paddles are connected to the Quick Disconnect (QD) assembly, and all 6 channels are operational.
- 6ch Neuro Flex1: Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 1 on the QD assembly and three channels are operational.
- 6ch Neuro Flex2: Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 2 on the QD assembly and three channels are operational.
- 3ch Neuro Flex1: Single paddle connected to Port 1 on the QD assembly, and three channels are operational.
- 3ch Neuro Flex2: Single paddle connected to Port 2 on the QD assembly, and three channels are operational.

Coil modes for general purpose applications:

- 6ch GP Flex: Both coil paddles are connected to the QD assembly, and all 6 channels are operational.
- 6ch GP Flex1: Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 1 on the QD assembly and three channels are operational.
- 6ch GP Flex2: Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 2 on the QD assembly and three channels are operational.
- 3ch GP Flex1: Single paddle connected to Port 1 on the QD assembly, and three channels are operational.
- 3ch GP Flex2: Single paddle connected to Port 2 on the QD assembly, and three channels are operational.

Page intentionally left blank

2

Chapter 2 Installation

2.1 Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance (MCQA) test

Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	60 minutes	-

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Short Loader Sphere for 1.5T (Green Solution)	1	2135650-2	-
Short Loader Cylinder for 1.5T	1	2135652-2	-
Head TLT Sphere for 3.0T (Pink Solution)	1	2359877	-
Head TLT Loader for 3.0T	1	2360031	-
Curved Adapter Panel for Flat Table (GEM)	1	5395828	Sumi Motherson

Replacement parts			
Item	Quantity	Part number	Manufacturer
Patient Security Strap (provided with system)	1	-	-

About this task

Coil configuration is part of the software build. Plug the coil into the system and make sure the coil is recognized by the system. If the applicable 6-channel flex coil is not configured for the site, refer to the latest revision of the *Coil Compatibility Matrix* (DOC1281278), available from the online documentation library.

Procedure

1. Before starting the MCQA test, the flex coil must be properly aligned. Put one coil half onto the table top.
2. Put the phantom with loader onto the coil half.

3. Put the second coil half on top of the phantom loader.

NOTE

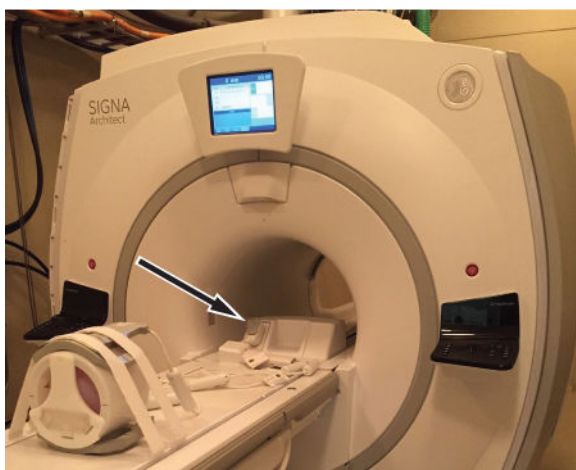
The flex coil cables should lay flat on the table top. Make sure these cables do not contain loops, such as those shown below.

4. Plug the flex coil Quick Disconnect (QD) into Port A on the LPCA. If the system does not have Port A, use the P to A adapter and plug it into P1 (the coil will not work in P2 or P4).

Figure 2-1 Flex coil connected to LPCA with looped coil



Figure 2-2 P to A adapter connected to P1



5. Set the landmark using the alignment lights with the crosshairs marked on the anterior half of the coil.
 - 5.1. At the magnet, press the **Alignment** button to turn on the laser alignment lights.

NOTE

Make sure the flex coil and phantom with loader are properly aligned (centered left to right) with the laser alignment lights.

Figure 2-3 Setting landmark on flex coil

- 5.2. Move the cradle to align the coil landmarks with the laser alignment lights.
- 5.3. Press the **Landmark** button to save the alignment.
6. Install the patient security strap (provided with the system) to secure both coil halves and the phantom with loader to the table top as shown.

NOTE

Make sure the security strap is properly aligned (centered front to back) on the flex coil and phantom with loader.

Figure 2-4 Flex coil and phantom with loader with patient security strap**Finalization**

Run the MCQA test. To do the MCQA test for a 1.5T or 3.0T system, locate the appropriate Service Methods procedure for the system. The *Multi-Coil Quality Assurance (MCQA) Tool* procedure is located in **Troubleshooting > System**.

Page intentionally left blank

Chapter 3 Replacements

3.1 Field Replaceable Units (FRUs)

The two coil halves of the 6-channel flex coil assembly are identical in design and manufacture. The coil halves are not matched pairs, and do not require tuning, so individual coil halves can be ordered and replaced.

Each coil half plugs into an intermediate adapter connector, otherwise known as a Quick Disconnect (QD) assembly, which is then plugged into the LPCA on the scanner. The QD assembly contains six imbedded pre-amps (one for each channel) that can be ordered separately.

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
3/16 Inch Flat Blade Screw-driver	1	-	-
1/8 Inch Flat Blade Screw-driver	1	-	-

NOTICE



POTENTIAL COMPONENT DAMAGE - ELECTROSTATIC DISCHARGE

Accumulated electrostatic charge can build up and discharge, causing damage to computer components.

Prevent damage to components by following Electrostatic Discharge (ESD) precautions:

- Work on a static-free mat.
- Wear a static strap to make sure that any accumulated electrostatic charge is discharged from your body to the ground.
- Create a common ground for the equipment you are working on by connecting the static-free mat, static strap, and peripheral units to that piece of equipment.
- For more information on ESD protection, see *MR Service Safety Manual* (5452735).

FRUs

Field Replaceable Units (FRUs) are listed below.

Description	Part number
1.5T 6-Channel Flex Phased Array Coil Half (each coil half is identical; therefore, a single half is permitted for replacement)	5229623
1.5T DV 6-Channel Coil Quick Disconnect Assembly	5363382
3.0T 6-Channel Flex Phased Array Coil Half (each coil half is identical; therefore, a single half is permitted for replacement)	5229646
3.0T DV 6-Channel Coil Quick Disconnect Assembly	5363382-3

Description	Part number
6-Channel Flex Coil Accessories Spares Kit (needed only when the coil is used on the Surgical Suite product; kit contains replacement parts to support the coil strain relief and mounting mechanism)	5269901
1.5T Low Noise Pre-Amplifier (channel location will be marked by the Field Engineer on the pre-amplifier with a permanent marker at the time of replacement)	2409869
3.0T Low Noise Pre-Amplifier (channel location will be marked by the Field Engineer on the pre-amplifier with a permanent marker at the time of replacement)	2409871
Short Loader Sphere for 1.5T (Green Solution)	2135650-2
Short Loader Cylinder for 1.5T	2135652-2
Head TLT Sphere for 3.0T (Pink Solution)	2359877
Head TLT Loader for 3.0T	2360031
P to A 6-Channel Flex Coil Adapter	5771497

3.2 Replacing the 6-channel flexible phased array coil assembly

About this task

The 6-channel flexible phased array coil assembly consists of two coil halves and the Quick Disconnect (QD) assembly. There are no replaceable parts within each of the coil half assemblies.

3.2.1 Replacing the 6-channel flex phased array coil half

About this task

The two coil halves of the 6-channel flexible phased array coil assembly are identical. Replace one coil half as needed.

Procedure

1. Make sure the 6-channel flexible phased array coil is moved out of the magnet room.
2. Disconnect the faulty 6-channel flexible phased array coil connector from the Quick Disconnect (QD) assembly.
3. Remove the replacement 6-channel flexible phased array coil from the shipping container.
4. Connect the new 6-channel flexible phased array coil connector to the QD assembly.
5. Put the coil removed from QD assembly into the shipping container.
6. Set up the flex coil and do the Multi-Coil Quality Assurance (MCQA) test. See [2.1 Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance \(MCQA\) test on page 17](#).

3.2.2 Replacing the Quick Disconnect (QD) assembly

Procedure

1. Make sure the 6-channel flexible phased array coil is moved out of the magnet room.
2. Disconnect the 6-channel flexible phased array coil first half connector from Port 1 of the QD assembly.

Figure 3-1 QD assembly - Port 1 and Port 2



1	Port 2
2	Port 1

3. Disconnect the 6-channel flexible phased array coil second half connector from Port 2 of the QD assembly.
4. Remove the replacement QD assembly from the shipping container.
5. Connect the 6-channel flexible phased array coil first half connector to Port 1 of the new QD assembly.
6. Connect the 6-channel flexible phased array coil second half connector to Port 2 of the new QD assembly.
7. Put the QD assembly removed from the coil halves into the shipping container.
8. Set up the flex coil and do the Multi-Coil Quality Assurance (MCQA) test. See [2.1 Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance \(MCQA\) test on page 17.](#)

3.2.3 Replacing the Quick Disconnect (QD) assembly pre-amplifiers

About this task

NOTICE



POTENTIAL COMPONENT DAMAGE - ELECTROSTATIC DISCHARGE

Accumulated electrostatic charge can build up and discharge, causing damage to computer components.

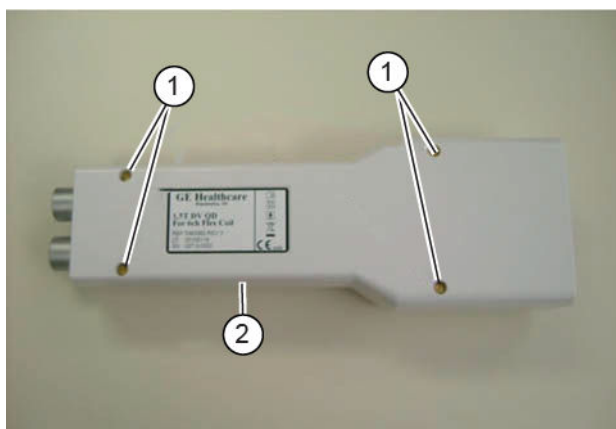
Prevent damage to components by following Electrostatic Discharge (ESD) precautions:

- Work on a static-free mat.
- Wear a static strap to make sure that any accumulated electrostatic charge is discharged from your body to the ground.
- Create a common ground for the equipment you are working on by connecting the static-free mat, static strap, and peripheral units to that piece of equipment.
- For more information on ESD protection, see *MR Service Safety Manual* (5452735).

Procedure

1. Move the QD assembly out of the magnet room.
2. Remove the four screws and bottom cover from the QD assembly.

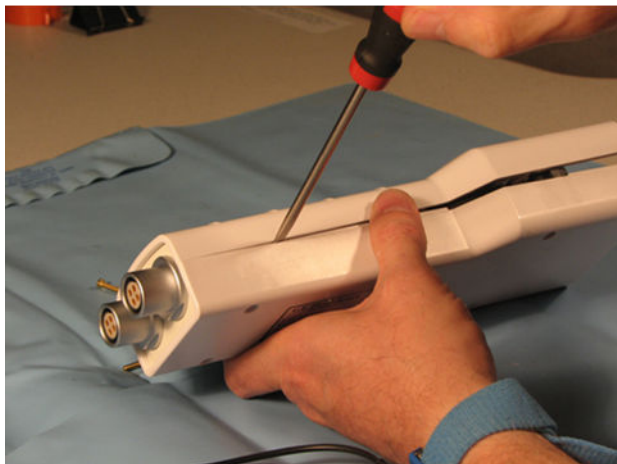
Figure 3-2 QD assembly - four screws and bottom cover



1	Screws
2	Bottom cover

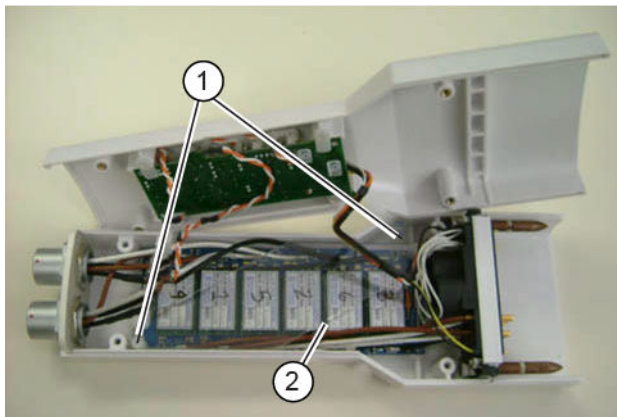
3. If necessary, use a 1/8 inch flat blade screwdriver and gently pry the housing halves of the QD assembly apart.

Figure 3-3 Separating housing on the quick disconnect assembly



4. Remove the two screws and Lexan cover from the QD assembly using a 3/16 inch flat blade screwdriver.

Figure 3-4 QD assembly - two screws and Lexan cover



1	Screws
2	Lexan cover

NOTICE



POTENTIAL COMPONENT DAMAGE

Pay special attention while removing the pre-amplifier from the socket to avoid damage to the printed circuit, components, or pre-amplifier.

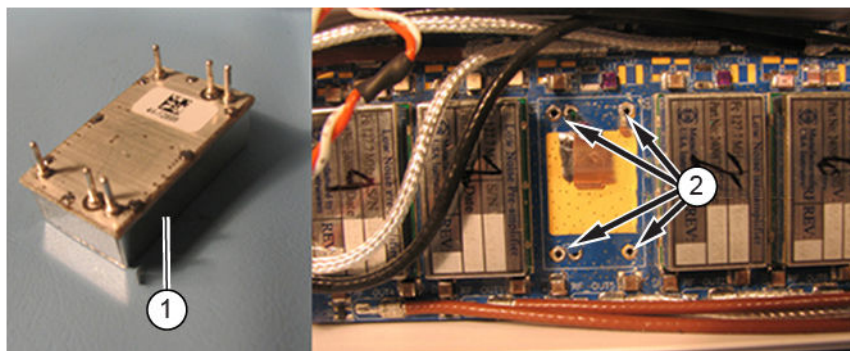
5. Remove the pre-amp from the QD assembly by prying under the pre-amp module on alternate sides of the module with a flat blade screwdriver.

Figure 3-5 Removing pre-amp from QD assembly



6. Refer to the figure below to identify the socket and pin locations for the pre-amp module.

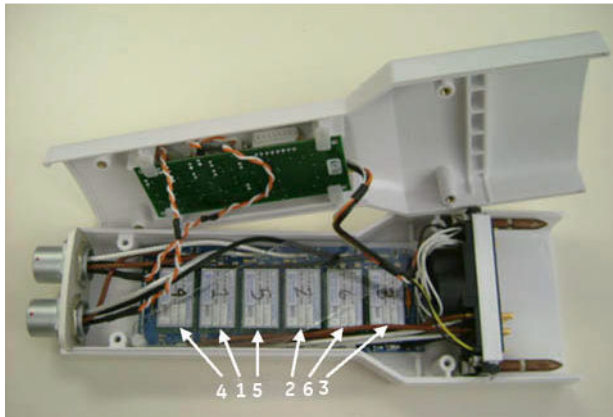
Figure 3-6 Socket and pin locations on the pre-amp



1	Socket
2	Pin locations

7. Make a note of the channel (location) number on the pre-amp so the new pre-amp can be marked with the same number.

Figure 3-7 QD assembly - pre-amp locations



8. Remove the replacement pre-amp from the shipping container and static bag.
9. Install the new pre-amp into the QD assembly, and use a permanent marker to record the channel number on the new pre-amp.

Figure 3-8 Marking channel number on QD assembly pre-amp

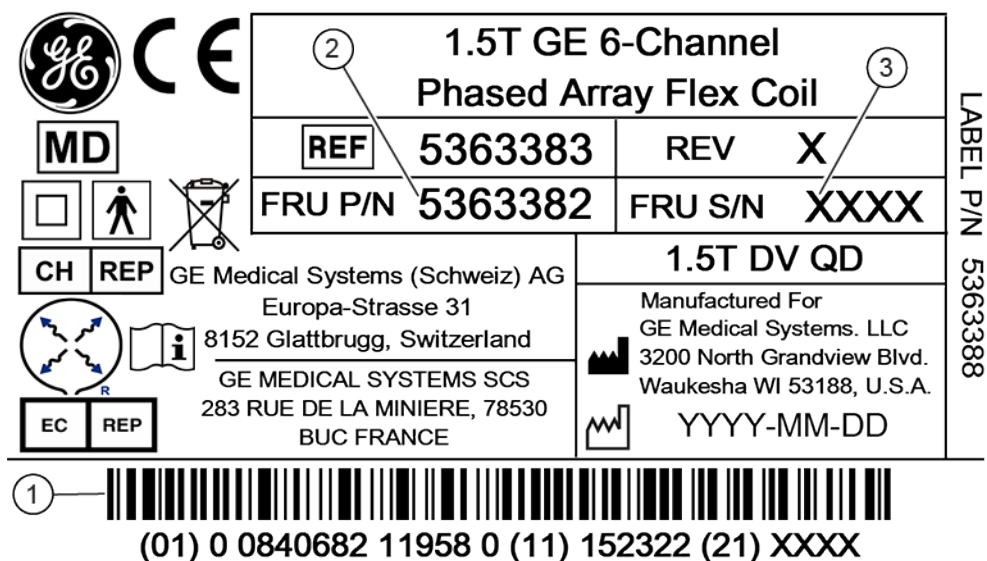


10. Install the Lexan cover on the QD assembly and fasten with two screws.
11. Put the pre-amp removed from the QD assembly into the static bag and into the shipping container.
12. Install the bottom cover to the QD assembly and fasten with four screws.
13. Set up the flex coil and do the Multi-Coil Quality Assurance (MCQA) test. See [2.1 Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance \(MCQA\) test on page 17.](#)

Finalization

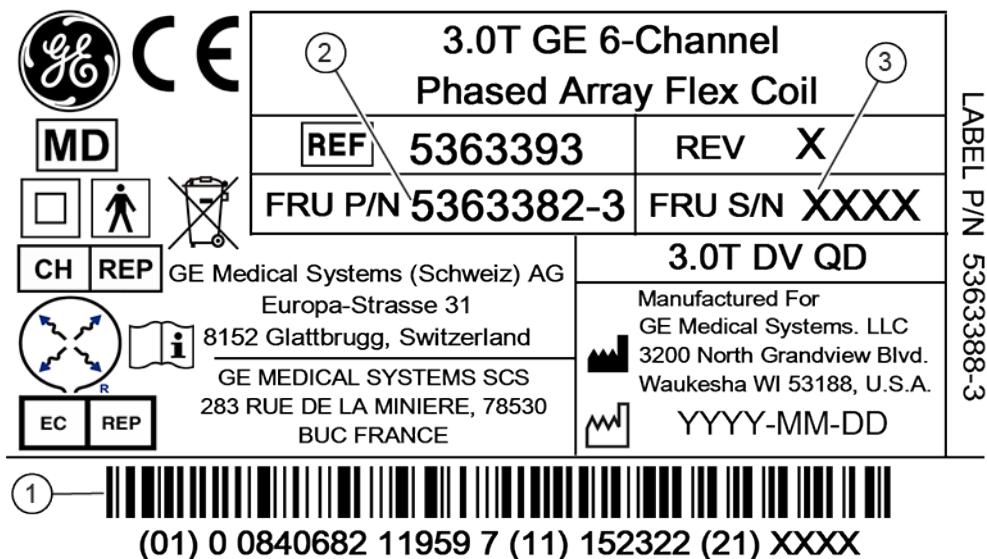
Update the Global Installed Base (GIB) with the new serial number information of the product locator for each new assembly replaced.

Figure 3-9 QD assembly - ID label example - 1.5T



1	UDI bar code
2	Part number
3	Serial number

Figure 3-10 QD assembly - ID label example - 3.0T



1	UDI bar code
2	Part number
3	Serial number

Chapter 4 Troubleshooting

4.1 Troubleshooting the 6-channel flex coil

The two coil halves of the 6-channel flexible phased array coil assembly are identical in design and manufacture. The foam sealed housing for the coil provides the required flexibility needed for neuro and general purpose imaging applications. This, however, restricts access to the internal components so troubleshooting the coil is done at the coil half assembly level. The coil halves are not matched pairs, and do not require tuning so individual coil halves can be ordered and replaced.

To determine if the problem lies in the coil or the receive chain, try swapping the coil cables at the Quick Disconnect (QD) assembly to see if the problem moves with the cable swap.

- If the problem moves, then the issue is with the coil half.
- If the problem does not move with the cable swap, use the MCRv Tool to determine where in the receive chain the problem lies.

Each coil half plugs into an intermediate adapter connector, otherwise known as a QD assembly, which is then plugged into the LPCA on the scanner. Imbedded pre-amps are provided in the QD assembly.

Hardware inspection

1. Inspect the foam housing of each coil half and look for cracks, discoloration, or other damage.
2. Inspect the cables and look for signs of damage such as cuts in the insulation, or kinks or twists, especially at the coil and connector entry points.
3. Inspect the cable connector and connector pins. Look for signs of damage to the pins such as corrosion, bent pins, or missing pins.
4. Make sure the field strength labeled on the coil is correct for the scanner on which it is used.
5. Inspect the QD assembly connectors for bent or missing pins.
6. Inspect the QD assembly housing for cracks, discoloration, or other damage.
7. Inspect the connectors on the LPCA and look for damaged or missing pins.
8. Make sure the coil cable connector is fully engaged into the QD assembly connector.
9. Make sure the QD assembly is fully engaged into the LPCA connector.

System inspection

1. Make sure that the scanner's operating frequency is within the bandwidth of the coil. See [1.1 Introduction on page 11](#) for the bandwidth specification for the coil. Corrections to system operating frequency may be necessary to improve coil performance.
2. Review the customer's images and look for shading or artifacts.

Multi-Coil Quality Assurance (MCQA) analysis report overview

The following illustrations depict the type of report resulting from a faulty coil connection.

Figure 4-1 Loss of signal on channel 5

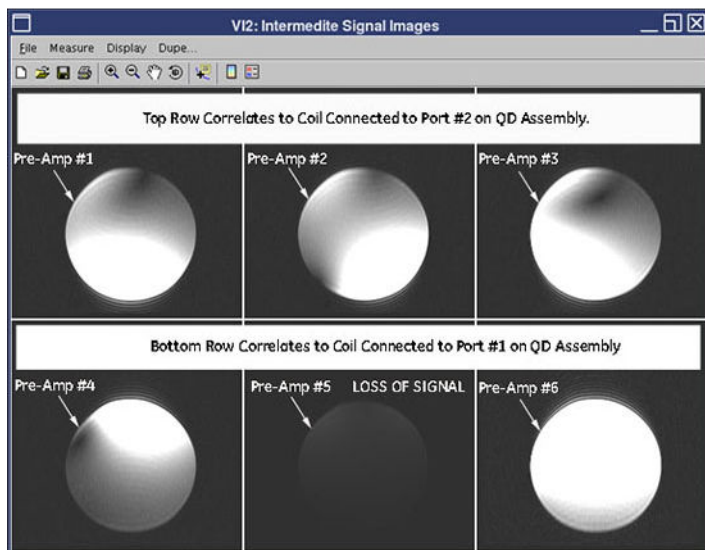


Figure 4-2 Loss of signal on channel 5

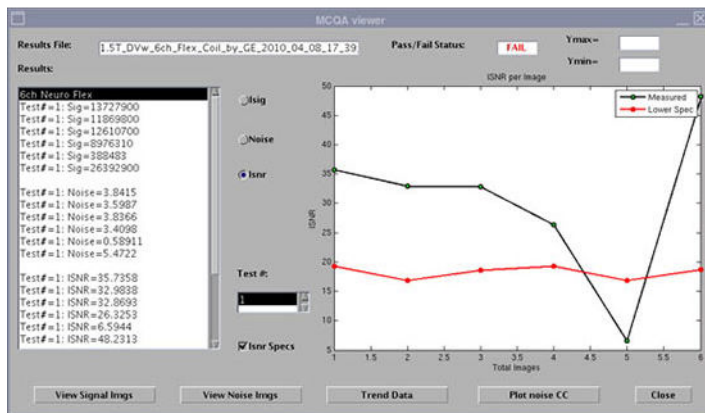


Figure 4-3 Loss of signal on channel 3

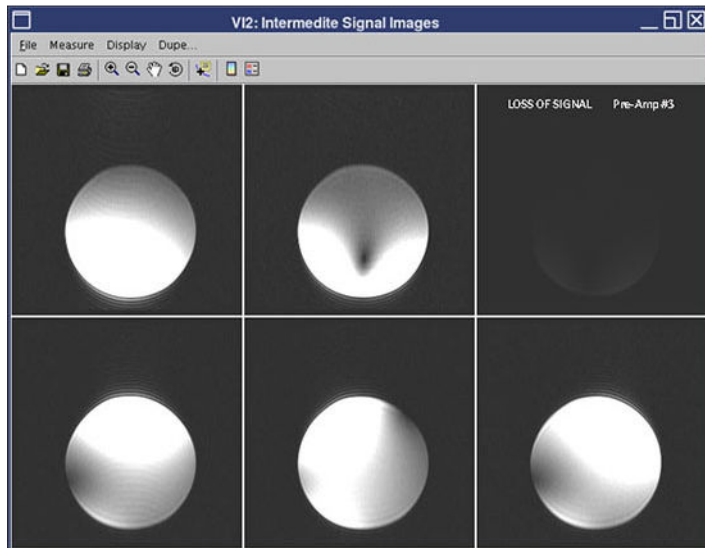
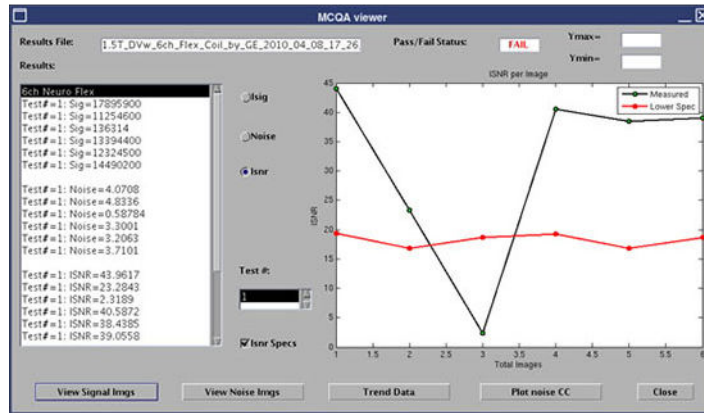


Figure 4-4 Loss of signal on channel 3

4.1.1 Troubleshooting the 6-channel flex coil using the multi-coil bias diagnostic

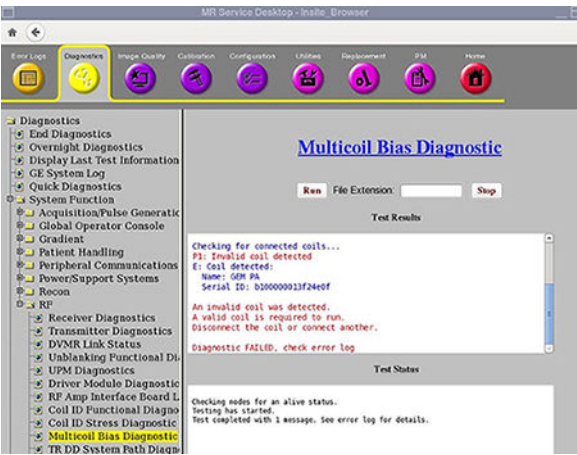
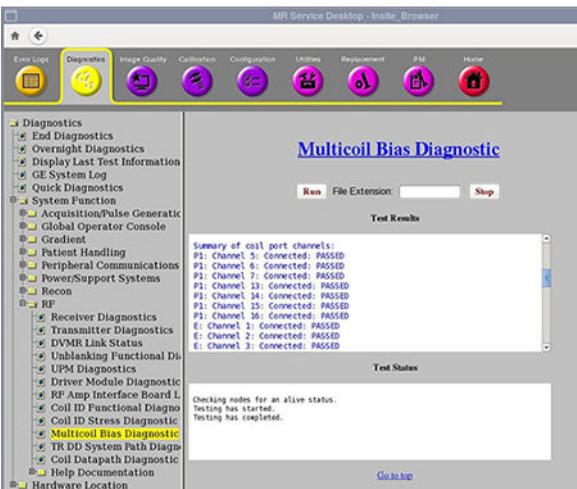
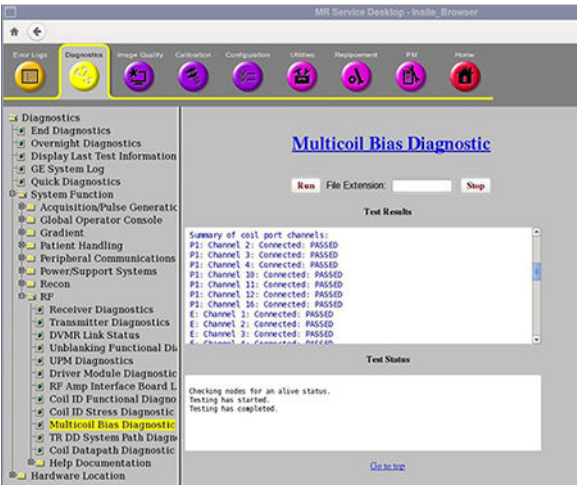
Expected multi-coil bias results on good parts

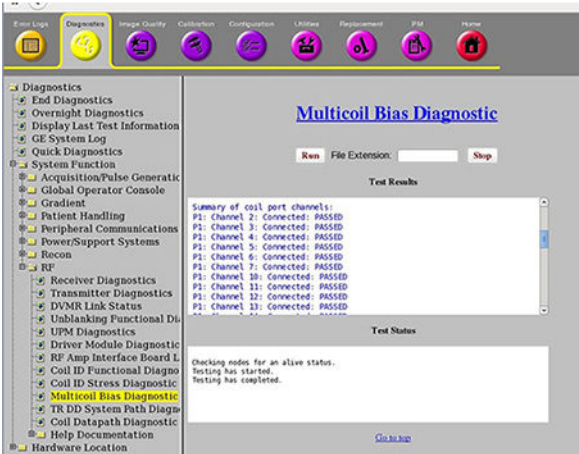
NOTE

Results differ for 1.5T and 3.0T.

The multi-coil bias diagnostic can be used to troubleshoot if the problem lies in the coil half or the Quick Disconnect (QD) box/P to A adapter. When the multi-coil bias is run with the identified parts, the following shall occur for a good part(s):

Plugged into P1	Result for good part(s)	Notes
Only the P to A adapter.		Multi-coil bias will detect a short on channel 16. This is by design and not a failure.

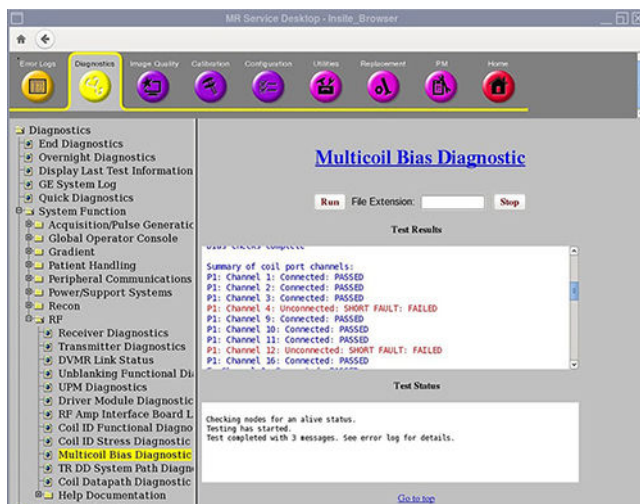
Plugged into P1	Result for good part(s)	Notes
P to A with QD box (no coil halves).	 NOTE 1.5T multi-coil bias results shown.	Multi-coil bias will not test in this configuration and will indicate Invalid coil in P1.
P to A with QD box and coil half plugged into QD box Port 1 only.	 NOTE 1.5T multi-coil bias results shown.	1.5T: Multi-coil bias tests only channels 5, 6, 7, 13, 14, 15, and 16 for coil in QD box Port 1. 3.0T: Multi-coil bias tests only channels 4, 5, 6, 12, 13, 14, and 16 for coil in QD box Port 1.
P to A with QD box and coil half plugged into QD box Port 2 only.	 NOTE 1.5T multi-coil bias results shown.	1.5T: Multi-coil bias tests only channels 2, 3, 4, 10, 11, 12, and 16 for coil in QD box Port 2. 3.0T: Multi-coil bias tests only channels 1, 2, 3, 9, 10, 11, and 16 for coil in QD box Port 2.

Plugged into P1	Result for good part(s)	Notes
P to A with QD box and coil half plugged into QD box Port 1 and Port 2.	 NOTE 1.5T multi-coil bias results shown.	1.5T: Multi-coil bias tests channels 2 through 7, and 10 through 16. 3.0T: Multi-coil bias tests only channels 1 through 6, 9 through 14, and 16.

Multi-coil bias results on a failed part

When there is a short or failure in the QD box, you will see a failure on two channels. In the example below, the test configuration is the P to A adapter, QD box with coil plugged into only port 2:

Figure 4-5 Short or failure in the QD box



Page intentionally left blank

Revision history

Revision	Date	Description
Controlled document for English is posted as DOC0757335.		
4	November 2023	- Applied SIMS style to existing content. - In <i>Chapter 1 Getting started</i>, section 1.1.1 <i>Shipping components</i>, <i>Table 1-1 Shipping contents by catalog number</i>: removed Coil Configuration CD (5372128) and 6-Channel Flex Coil Service Pack DVD (5785805). - In <i>Chapter 2 Installation</i>, section 2.1 <i>Setting up the 6-channel flex coil for the Multi-Coil Quality Assurance (MCQA) test</i>: - In the <i>Tools and test equipment</i> table: updated part number 2135650 to 2135650-2 and part number 2135652 to 2135652-2, and updated descriptions. - Added <i>About this task</i> section. - Removed <i>Coil Configuration</i> section. - Updated <i>Finalization</i> section to remove steps and update topic location. - In <i>Chapter 3 Replacements</i>: - In section 3.1 <i>Field Replaceable Units (FRUs)</i>, <i>FRUs</i> table: updated part number 2135650 to 2135650-2 and part number 2135652 to 2135652-2, and updated descriptions. - In section 3.2.3 <i>Replacing the Quick Disconnect (QD) assembly pre-amplifiers</i>, <i>Finalization</i> section: - Replaced <i>Illustration 3-9: Quick Disconnect - ID Label Example</i> with <i>Figure 3-9 QD assembly - ID label example - 1.5T</i>. - Added <i>Figure 3-10 QD assembly - ID label example - 3.0T</i>.
3	September 2017	Updated for addition of P to A adapter 5771497.
2	May 2011	Updated for MR450w GEM.
1	July 2010	Initial release.

Page intentionally left blank

Released

Released



www.gehealthcare.com